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The role of AI in (Re)Shaping energy Finance: A systematic literature review

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ABSTRACT

Energy finance is an interdisciplinary, rapidly evolving field integrating finance, economics, and energy systems to address complex challenges in market dynamics, sustainability, and policy-making amidst global energy transitions to net zero. Recent advances in artificial intelligence have reshaped energy finance research and practice, but no prior study has systematically examined how these methods have been applied across the field or what value they have brought. This paper provides a comprehensive review of approximately 700 academic studies published over the past five years, analyzing the diffusion and contribution of AI techniques across the six established thematic areas proposed by Zhang et al. (2018). While machine learning models are increasingly adopted, especially in forecasting applications, and often outperform classical econometric approaches in terms of accuracy, their use remains uneven. In some cases, methodological choices are clearly adapted to the structure of the data and the objective of the study, while in others, the implementation of complex ML algorithms lacks contextual justification or interpretability. More importantly, the review highlights a major gap, namely recent approaches such as explainable AI and causal machine learning are almost entirely absent, despite their potential to enhance transparency, support causal inference, and inform energy-related financial decisions. By identifying where AI has already contributed, where it remains underused, and how it could be better integrated, the paper offers a structured and forward-looking perspective on the evolving relationship between artificial intelligence and energy finance.

1. Introduction

Energy is a strategic input to the global economy. Despite international efforts to increase energy efficiency and decouple economic growth from energy consumption, energy demand is rising quickly. In this context, energy products are increasingly viewed as financial assets, leading to a greater interconnection between financial and energy markets. Geopolitical turmoil, particularly the Russian invasion of Ukraine, has intensified the global energy crisis and highlighted the vulnerability of energy systems. Regulatory changes to raise resilience and improve energy risk management, alongside long-term governmental initiatives to enforce more efficient and sustainable resource management, attain net zero, and combat climate change, have further strengthened the link between energy and financial markets. These developments demand a holistic approach toward energy finance.

Much literature emphasizes the interdependency of energy and financial markets, whether presenting energy markets as a source of financial shocks, highlighting the growing financialization of energy

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products, or exploring the role of green investments in shaping societal development. However, despite this growing body of work, the research landscape remains complex and fragmented, lacking a coherent structure that integrates technological, financial, and policy dimensions. The seminal work of Zhang et al. [1] made a notable attempt to systematize the field by identifying six core domains, namely Energy and Financial Markets, Pricing Mechanism, Energy Corporate Finance, Green Finance and Investment, Energy Derivative Markets, and Energy Risk Management. These categories serve as the structural foundation for the present study. Zhang et al. [1] also emphasizes the importance of quantitative methods in energy finance but does not anticipate the profound methodological shift brought about by artificial intelligence. Our review takes this structure as a starting point, without seeking to revise or expand it. Instead, we focus on how artificial intelligence, an aspect not covered by Zhang et al. [1], has been applied across these domains, identifying where it has already made a difference and where it remains underutilized.

In recent years, the rapid advancement of AI has begun to reshape these domains in fundamental ways. As the capacity to process and learn from large volumes of data has expanded, AI has increasingly influenced how energy and financial systems are modeled and managed. These techniques have contributed to improved forecasting accuracy, greater operational adaptability, enhanced integration of renewable sources, and more robust responses to uncertainty. Since 2022, this intersection has attracted intense academic interest, catalyzed by the dual shock of the COVID-19 pandemic and the war in Ukraine. Over ninety-five percent of AI-related studies in energy finance have been published after 2020, reflecting not only a shift in methodological preferences but also a broader transformation in how energy resilience, investment strategy, and regulatory mechanisms are conceptualized and analyzed through data-driven approaches. Although artificial intelligence is now widely used across various sectors of the economy, no existing review has systematically analyzed how AI techniques have been applied within the energy finance literature. Studies such as Entezari et al. [2] and Ghoddusi et al. [3] have explained this intersection only marginally and do not provide a structured analysis of AI methods across the six key areas of energy finance, nor do they compare the performance of AI models with classical econometric approaches.

This review addresses several critical gaps in the existing literature on energy finance. While previous surveys have acknowledged the growing role of AI, none have systematically examined how these methods have been deployed across the main empirical areas of the field, nor have they assessed whether their application reflects methodological fit, data adequacy, or a genuine improvement over traditional econometric approach. By covering almost 700 peer-reviewed articles along the six-pillar framework proposed by Zhang et al. [1], this paper provides, up to our knowledge the first structured mapping of AI adoption in energy finance, capturing both the intensity and purpose of different techniques. It disaggregates AI into core methodological families and demonstrates that, while models such as LSTM, SVM, and tree-based ensembles dominate forecasting and risk management, their use in corporate finance and green investment remains limited, often lacking theoretical justification in terms of panel data or interpretability. The analysis confirms that machine learning techniques are increasingly preferred over traditional econometric models, especially in forecasting tasks where models must handle large volumes of daily data and capture complex, nonlinear relationships. Certain methods, such as ensemble algorithms and neural networks, are predominantly applied to oil and natural gas price prediction, reflecting a strong methodological focus on short-term market dynamics. At the same time, explainable artificial intelligence and causal machine learning remain almost entirely absent from the literature, despite their clear potential to enhance transparency, support policy evaluation, and address issues of distributional impact and fairness. In response, this review outlines several research directions that deserve further attention, including the integration of causal inference tools in the evaluation of energy pricing, subsidies, and transition policies, the use of interpretable models in regulatory and investment decision-making, and a more critical assessment of when and where complex AI methods offer practical value over simpler alternatives.

2. Methodology/PRISMA

2.1. A brief recap of the PRISMA approach

We follow the PRISMA guidelines in their most recent version to ensure our survey's completeness, transparency, and reproducibility [4]. The main tools of PRISMA are the PRISMA checklist, which contains 27 recommendations, and the PRISMA flow diagram, which shows a numerical synthesis of the sources considered for inclusion and finally included in the review paper. We create a single PRISMA flow diagram for the six energy finance dimensions to document the number of studies identified, screened, included, and excluded and the reasons for exclusion.

2.2. How do we instantiate PRISMA

Zhang [1] defines the energy finance pillars as follows. The field of *Energy and financial markets* explores how energy price shocks affect stock returns. *Pricing mechanisms* analyze how energy prices are formed, influenced not only by supply and demand but also by the financialization of energy commodities. *Energy corporate finance* focuses on energy firms' specific financing and investment decisions. *Green finance and investment* highlight the importance of financing sustainable energy projects and developing emissions trading schemes. *Energy derivative markets* examine the role of derivatives in price discovery and managing energy price volatility. *Energy risk management* discusses strategies to hedge against market risks and maintain energy security. We organize our review around these pillars. This way, we further elaborate on their definition by analyzing many recent papers. Moreover, we review their topicality in light of recent advancements in AI techniques and the escalating challenges in the energy market, such as the European energy crisis, the war in Ukraine, climate change, and the increasing demand for sustainable energy solutions.

We rely on the Web of Science (WoS) platform, one of the most reputable and widely cited databases for academic papers. We also used a flexible PRISMA approach to cope with the differences between the pillars. More specifically, we added a supplementary layer for some pillars to increase the relevance of the identified papers.

We seeded the stepwise PRISMA approach for each of the six energy finance pillars with a list of keywords proposed by Zhang [1]. *The complete list of words for this initial layer is available in* Appendix 2. The second step represented a combined layer of forecasting and AI-related keywords, which we defined based on an extensive literature review. *The list of keywords is also available in* Appendix 2. The third step involved using specific keywords related to each of the six pillars to narrow down the results and raise their relevance. The fourth step comprises a qualitative filter based on the journal where an article has been published. Only articles published in peer-reviewed journals from the first and second quartile were selected. Finally, we filtered papers considered loosely or not at all connected to the specific pillar for which they were initially selected. Using a five-step approach for each pillar, we compiled and analyzed a list of papers to establish the state-of-the-art and identify the key characteristics of each energy finance pillar. Table 1 reports the number of studies selected/filtered at each stage.

3. Energy finance state-of-the-art

As a continuation of the methodological approach, it is worth emphasizing how naturally energy finance connects with the use of artificial intelligence. Energy markets are complex and constantly changing, and the data available has become more detailed and more

Table 1
Summary of the results of the review process.

Identification of studies through databases and registers (Subtopics as defined by Zhang et al. (2018))	Energy and financial markets	Pricing mechanisms	Energy corporate finance	Green Finance and Investment	Energy derivatives market	Energy and risk management
Records identified in WoS	**101**	**103**	**31**	**231**	**65**	**117**
Records removed before screening	0	0	0	0	0	0
Records added[a]	0	0	16	0	0	0
Duplicate records removed	0	0	0	0	1	0
Records marked as ineligible by automation tools	0	0	0	0	1	0
Records removed for other reasons – lack of AI-related content	0	0	0	50	6	0
Articles screened	**101**	**103**	**47**	**171**	**57**	**117**
Articles excluded**	63	83	17	123	6	74
**Not relevant to the main focus or topic relevant	52	27	17	103	6	51
**Not Q1 or Q2	11	56	0	20	0	23
Articles included in this review	**40**	**20**	**30**	**48**	**51**	**43**

[a] We carefully reviewed each of the selected articles to identify other relevant studies that might have been overlooked by Web of Science queries. Additionally, we checked the citations available on Web of Science for these articles to identify potential research directions conducted using machine learning (ML) techniques.

frequent. In this environment, researchers are increasingly turning to AI tools to improve forecasting accuracy, better understand market risks, and evaluate the effects of policies in a more dynamic way. While AI methods were initially used as a complement to classical models, they now play a central role in many applied studies. As a result, the literature shows a wide range of approaches, including decision trees, variable selection techniques, neural networks, probabilistic models, and methods that work with text data. Given this diversity, it no longer makes sense to treat AI as a single, uniform method. It is important to understand the main families of AI techniques, why they are used in specific contexts, and what kinds of problems they are best suited to solve. Table 2 provides an overview of these methods and how they are applied in the literature on energy derivative markets.

Moving forward, in order to better understand how the literature has evolved and where it currently stands, we provide a consolidated view of the key patterns emerging across the six pillars of energy finance.

Table 2
Classification of AI techniques.

AI Methodological Family	Purpose or Typical Features	Application in Energy Derivative Markets
Tree-based methods (for example Random Forest, Gradient Boosting Machines)	These models are used to capture nonlinear relationships, reduce overfitting, and identify important predictors through structured splitting rules and ensemble averaging.	They are frequently used in the literature for forecasting price movements and modeling volatility in energy markets.
Bayesian or probabilistic models	These models allow estimation under uncertainty by producing probability distributions over outcomes. They are useful for stress scenarios and risk quantification.	These methods are used for prediction tasks under uncertainty, especially in contexts involving market stress or instability.
Neural networks (for example Multilayer Perceptrons, Recurrent Neural Networks, Convolutional Neural Networks)	Neural networks are applied to recognize patterns and make predictions from complex datasets, especially those with temporal, sequential, or multivariate structure.	They are widely applied for short-term forecasting and detecting patterns in high-frequency or multi-dimensional data related to energy products.
Natural Language Processing	Natural Language Processing techniques help extract meaningful signals from unstructured text data such as market news, reports, or commentary.	There are a few studies that apply Natural Language Processing to extract sentiment or relevant information from textual data in energy market contexts.

Table 3 organizes these insights by capturing the dominant types of research, the energy sectors most often studied, the geographic focus of the data, and the methodological approaches used. Special attention is given to the adoption of artificial intelligence, not only in terms of overall presence, but also regarding the specific techniques applied, such as neural networks, natural language processing, and other machine learning tools. Moreover, we track the extent to which newer developments like Causality in the classical sense and Explainable AI have been integrated into the literature, particularly in contexts where transparency and policy relevance are essential.

Thus, the first column represents the distribution of various types of research within the literature. For each subfield, we determined the percentage of identified articles that were reviews, benchmarking studies, methodological papers, or policy-oriented papers. This column illustrates the trend in the recent literature on energy finance and the possible gaps that targeted studies could cover. The second element analyzed is related to the energy market sector and presents, for each subfield, the percentage of articles studying the oil, gas, or electricity sector. This column illustrates whether research in a subfield concentrates on specific energy market sectors. The third column refers to the geographical regions. It shows the weight of papers that were carried out based on data for critical economic actors at the global level. For example, it considers the G7, the EU, BRICS countries, and other actors important for the energy market. The forth column presents the percentage of papers that use econometric techniques. The role of this column is to examine whether econometric methods remain the primary vehicle for quantitative analysis in a subfield and whether there are subdomains for which other types of analysis methods are more frequent. The fifth column presents the percentage of AI-oriented papers in each pillar. The purpose is to illustrate the degree of AI utilization. Likewise, the sixth column represents the most common types of AI techniques used in the literature. Specifically, we report the percentage of papers that use (Artificial) Neural Network (NN)-type techniques, Natural Language Processing (NLP)-type techniques, and other types of AI or Machine Learning (ML) techniques. This analysis helps to identify potential research gaps from a methodological point of view. The last column identifies the percentage of papers that use Causality and Explainable AI (XAI), two of the techniques known in the literature as relevant to the decision-making process [5]. These methods can make AI-based approaches a powerful instrument to shed light on policy-related research questions traditionally pursued using econometric/statistical techniques. Thus, the column clarifies whether a subfield embraces the latest family of AI-tools.

In what follows, we examine in more detail the most relevant contributions within each of the six pillars, with a particular focus on how artificial intelligence has shaped the empirical and methodological evolution of the field. To guide this assessment, we provide a

Table 3
Energy finance Subsections features.

Energy finance - subsections	Most common type of paper (% of total)			Target sector (% of total)			Data focus % of papers focusing on:		% of papers using econometric techniques	AI-oriented papers (% of total)	Most used AI techniques (% of total papers using AI)			% of papers looking at	
	Theoretical	Methodology/ Benchmarking	Policy	Oil	Gas	Electricity	EU and G7	BRICS			NN	NLP	Another AI/ML	Causality	XAI
A. Energy and Financial Markets	**2,5**	**85,00**	**72,50**	**62,50**	**5,00**	**35,00**	**75,00**	**22,50**	**100,00**	**97,50**	**62,50**	**7,50**	**30,00**	**47,50**	**2,50**
B. Pricing Mechanism	**5,00**	**95,00**	**45,00**	**0,00**	**5,00**	**85,00**	**45,00**	**35,00**	**25,00**	**100,00**	**80,00**	**0,00**	**20,00**	**10,00**	**5,00**
C. Energy Corporate Finance	**0,00**	**100,00**	**100,00**	**N/A**	**N/A**	**N/A**	**36,66**	**80,00**	**100,00**	**0,00**	**0,00**	**0,00**	**0,00**	**100,00**	**0,00**
D. Green Finance and Investment	**0,00**	**66,67**	**97,91**	**31,25**	**33,34**	**85,41**	**47,91**	**62,5**	**58,33**	**60,41**	**41,66**	**6,25**	**52,09**	**58,33**	**0,00**
E. Energy Derivative Markets	**0,00**	**100,00**	**3,92**	**94,11**	**19,6**	**1,96**	**82,35**	**29,41**	**43,13**	**100,00**	**88,23**	**11,76**	**11,76**	**1,96**	**19,6**
F. Energy Risk Management	**4,65**	**97,67**	**27,9**	**30,23**	**25,58**	**60,46**	**67,44**	**30,23**	**62,79**	**93,02**	**72,09**	**2,32**	**25,59**	**6,97**	**4,65**

Note: N/A indicates "Not Applicable."

comparative overview that summarizes key structural features of the literature associated with each pillar.

3.1. Energy and financial markets

From a methodological perspective, most of the selected studies evaluate the extent to which machine learning models improve forecasting accuracy in comparison to traditional econometric approaches. Sadorsky [6] applies random forest algorithms to predict the direction of clean energy stock prices using a set of technical indicators. Accuracy rates reach between 85 and 90 percent for a twenty-day forecasting horizon, substantially higher than the 55 to 60 percent obtained using logit models. Zhang et al. [7] address aggregate oil market volatility using a macroeconomic database of 127 variables and combine variable selection techniques such as lasso and elastic net with supervised principal component analysis. The resulting common factor index consistently improves forecasts in both statistical and economic terms, with option-implied volatility emerging as the most powerful predictor. Lu et al. [8] conduct a comprehensive comparison across traditional, machine learning, and combination models for oil futures volatility. Ensemble methods and neural networks yield the most consistent performance across different forecasting conditions, with combination models providing additional robustness.

Furthermore, Wang and Zhao [9] use Bayesian networks to identify macroeconomic and energy-related drivers of carbon prices under the European Union Emissions Trading System. Their model captures both direct and indirect transmission mechanisms through which equity and energy markets affect carbon futures. Xu and Wang [10] improve the reliability of empirical mode decomposition by introducing a rolling window structure and testing two non-iterative machine learning algorithms, random vector functional link and extreme learning machine, for long-term crude oil price forecasting. Results based on Brent and West Texas Intermediate spot prices indicate that the rolling denoising model outperforms other benchmark models. Ciner and Karagozoglu [11] integrate a textual climate-related uncertainty index into a volatility model for clean energy portfolios. Forecasting accuracy improves significantly, particularly in periods characterized by heightened uncertainty. Loutfi [12] develops a sentiment-based forecasting model using environmental television news transcripts from multiple networks. Sentiment scores, especially those derived from conservative outlets, enhance the forecast accuracy of renewable energy stock prices across multiple evaluation metrics. Mohsin and Jamaani [13] introduce a deep convolutional neural network to predict oil price volatility using the historical prices of five precious metals. The proposed model, called GroupNet, achieves higher accuracy than a set of machine learning and statistical benchmarks. Li et al. [14] combine deep learning with online media text mining and use latent Dirichlet allocation to group sentiment features by topic. The resulting forecasting models outperform earlier benchmarks by capturing the joint contribution of text-based and financial predictors.

All in all, these studies demonstrate that machine learning models offer substantial advantages when applied to forecasting tasks in energy-related financial markets. They usually outperform traditional econometric approaches in modeling volatility, predicting the direction of asset prices, and selecting relevant predictors from high-dimensional datasets. The integration of textual sentiment, macroeconomic signals, and market-based indicators allows these models to capture complex relationships that are often overlooked in linear frameworks. Forecasting accuracy improves not only in statistical terms but also in terms of economic relevance, especially when dealing with uncertainty, media influence, and the interaction between energy and carbon markets. Additional details on Energy and Financial Markets can be found in Appendix 1A.

3.2. Pricing mechanisms

From a methodological standpoint, recent research within the pricing mechanism domain increasingly emphasizes the role of machine learning in enhancing decision-making under uncertainty. Talluri et al. [15] develop a layered battery scheduling strategy for renewable energy communities, combining machine learning forecasts, MILP-based optimization, and intra-hour real-time adjustments. The model is calibrated under the Italian regulatory framework and delivers higher prosumer revenues while reducing stress on storage infrastructure. Building on this approach, Kim et al. [16] introduce a reinforcement learning algorithm to manage battery discharge patterns in the presence of time-of-use tariffs. Their framework minimizes operating costs while explicitly incorporating battery aging, a feature that strengthens its practical relevance in long-term energy storage management.

On the demand side, several studies explore how learning algorithms can improve household energy management and adaptive pricing strategies. Chellamani et al. [17] apply K-means clustering to define comfort intervals for schedulable appliances and use a nature-inspired optimization routine to reduce daily electricity bills. In a comparable context, Lee and Choi [18] propose a reinforcement learning system that coordinates smart appliance usage and battery operations, assisted by neural network-based forecasts of indoor temperature. These personalized systems are complemented by Salazar et al. [19], who combine reinforcement learning with customer segmentation to design dynamic pricing and incentive schemes. The results indicate substantial improvements in grid responsiveness and user satisfaction, suggesting that integrating consumer behavior into algorithmic pricing can enhance both efficiency and acceptability.

The interplay between user heterogeneity and pricing effectiveness is further explored through clustering-based demand-side management frameworks. Wen et al. [20] and Zhao et al. [21] propose approaches that segment consumers based on energy use patterns and socio-demographic features, allowing for differentiated pricing strategies such as real-time pricing and multistep tariffs. Their findings highlight the economic benefits of moving beyond flat-rate schemes. At a broader market level, Alova & Caldecott [22] develop a multi-output learning model to examine how environmental policy and market structures shape investment dynamics in renewable technologies. Their results point to significant differences between utilities and independent producers, with the latter responding more strongly to targeted subsidies and carbon pricing schemes.

In parallel, machine learning is increasingly used to improve the accuracy and interpretability of energy and carbon price forecasts. Enrich et al. [23] leverage ML-based variable selection to refine the identification strategy in a difference-in-differences framework, improving causal inference regarding time-of-use pricing in Spain. Lin et al. [24] adopt a deep hybrid forecasting model that combines signal decomposition, convolutional filtering, and BiLSTM architectures to predict spot electricity prices. Li et al. [25] extend this architecture to carbon pricing using a causal temporal convolutional network, offering insights into both predictive performance and structural causality. Complementary perspectives are provided by Herrera et al. [26], who compare traditional econometric and ML approaches for forecasting global commodity prices, and Ferrari et al. [27], who implement sparse dynamic factor models on macroeconomic panels to predict energy prices over longer horizons. Collectively, these studies demonstrate that ML-based forecasting and optimization tools not only enhance accuracy but also support more informed decision-making for producers, consumers, and regulators operating under volatile and complex pricing environments. Additional details on *Pricing Mechanism* can be found in Appendix 1B.

3.3. Energy corporate finance

The contribution to the literature of this pillar is rather policy-oriented. For example, from an investment behavior and efficiency perspective, Shi [28] suggested that free cash flow leads to over-investment in energy companies, indicating an inefficient use of resources. On the contrary, Lee and Xiao [29] suggest that supply chain financing improves the efficiency of investments in new energy enterprises. In terms of corporate governance, Kong et al. [30] emphasize the positive influence of foreign institutional investors on the innovation of energy companies, while Bu et al. [31] demonstrate that disclosure of environmental information enhances the energy efficiency of companies. Regarding the impact of environmental regulations and policies, Ren et al. [32] find that the uncertainty of oil prices reduces excessive corporate debt, while Deng et al. [33] demonstrate that environmental protection tax reform in China enhances the resilience of energy companies and reduces carbon emissions. Ultimately, research on risk and profitability underscores the need for digital finance and market risk management. Wu et al. [34] suggest that digital financial innovation can improve the financial performance of energy companies, while Apergis [35] and Li et al. [36] emphasize the need for strong risk management techniques to handle market volatility.

Against this background, we believe that the literature in the field of Energy Corporate Finance would greatly benefit from integrating ML techniques. Combining ML with standard panel data, econometric approaches such as panel fixed effects or System GMM has the potential to enhance research activities. ML approaches can improve the forecasting accuracy of these econometric models, identify challenging non-linear correlations, and handle large datasets with multidimensional characteristics more effectively [37]. For example, ML/AI may be used to capture different characteristics from big datasets while also identifying the most important variables for inclusion in econometric models. Moreover, Lasso regression may be used to choose variables in ways that standard econometric approaches cannot [38]. Furthermore, Random Forests or NN can reveal complicated, nonlinear patterns in data that typical panel data approaches may overlook. These patterns can be used to guide the implementation of econometric models, resulting in more accurate and robust outcomes.

In addition, incorporating ML into traditional panel models reduces the risk of model misspecification by automatically detecting complex patterns without using standard econometric hypothesis testing, which has often failed to fit the empirical data accurately, while also improving generalization capabilities through techniques like regularization and cross-validation. Finally, several papers included in this pillar use propensity score matching or difference-in-difference approaches to capture the causal patterns in observational data, so it is likely that, in the near future, advancements in Causal ML will enhance the quality of the results regarding different topics of energy corporate finance. As an example, AI techniques could be used to identify complex investment and efficiency patterns in energy company behavior, providing a deeper understanding of resource allocation and the impact of abundant financial flows, as shown by Shi [28] and Zhang and Kong [39]. Moreover, recent AI applications in areas such as churn prediction Jiang et al., [40] and environmental monitoring Dong et al., [41] offer methodological innovations that could inspire future work. Techniques like profit-driven feature optimization or novel metaheuristics for weighting ensemble models may prove useful in modeling energy-related behaviors, especially under data constraints or when targeting heterogeneous users. Additional details on Energy Corporate Finance can be found in Appendix 1C.

3.4. Green finance and investment

Regarding this direction of research, some empirical studies such as Lei et al. [42], Li et al. [43], Wang et al. [44], and Xu et al. [45] have proposed advanced carbon price forecasting methods, while Alova & Caldecott [22] use machine learning to assess the resilience of gas and coal to carbon pricing and the effectiveness of renewable subsidies. These studies highlight the crucial role of carbon pricing in the energy

transition, indicating that current prices are insufficient to drive significant technological changes in polluting industries [46]. Furthermore, other research papers emphasize the discrepancy in carbon emission efficiencies between developed and developing countries [47] and the varied impact of economic development, energy consumption, and clean energy on carbon prices. The United States is noted for its minimal explicit carbon pricing, with calls for more ambitious policy reforms to address this gap [48,49].

Until now, the field remains a relatively immature but interdisciplinary research area [50]. Green finance supports the development of renewable energy primarily through stimulating investment in fixed assets and encouraging innovation in clean technologies [51]. Ding et al. [52] demonstrate that financial development (FD) can improve the net-zero transformation potential (ENTP) of urban energy systems in China, although this depends heavily on city size, economic structure, and resource endowment. Zhou et al. [53] argue that accelerated and diversified green investments are essential for meeting climate targets under stringent policy scenarios in both Europe and China. Several articles in our selection confirm a causal relationship between green finance and the transformation of energy systems, with most of the evidence still concentrated in Chinese case studies [51,54–56]. Moreover, green finance has been shown to positively impact clean energy yields, stimulate local green economic development [56], and accelerate the deployment of wind, hydro, and solar generation [57].

However, the effects of green finance are far from uniform. Cross-country comparisons reveal major differences in how policies and market incentives influence technology choices, particularly between OECD and emerging economies. Green finance tends to be more effective in countries with strong regulatory frameworks, deeper financial markets, and ambitious climate targets. Conversely, developing economies with limited institutional capacity and weaker green financial infrastructure see more modest impacts [51,58]. In such contexts, the renewable energy sectors most dependent on external financing appear to benefit the most from targeted green finance instruments.

The literature also explores the policy implications of green finance and investment more broadly. Contributions range from studies analyzing the pass-through of carbon costs to wholesale electricity prices in Japan [59], to meta-analyses of the most commonly used methods in energy economics research [3]. Coordinated policy frameworks are increasingly recognized as critical for aligning energy, financial, and environmental goals [60]. While Martinez-Alvarez et al. [61] argue that political leadership plays a limited and often transitory role in energy outcomes, other work suggests that effective policy coordination can accelerate system-level change. The influence of the European Union Emissions Trading System (EU ETS) since 2005 is a recurring theme, particularly its role in integrating carbon, energy, and financial markets [62]. Additional details describing Green Finance and Investment can be found in Appendix 1D.

3.5. Energy derivative markets

The extensive research on this topic, with examples including [63,64] and Yang et al. [65], among others, which specifically investigated the improvement of recurrent neural network (RNN) methodologies through the application of stochastic random-time functions, underscores the necessity for sophisticated regulatory frameworks. The enhanced predictive precision of these models, compared to conventional econometric and ML models, might assist policymakers in anticipating market shifts more effectively and proposing energy policies that are more robust. Innovative causality tests suggested by Creamer and Lee [66] emphasize the need for regulatory agencies to implement more advanced analytical techniques to properly handle energy market concerns. Furthermore, research by Shaikh [67] reveals that utilizing the Oil Volatility Index (OVX) as a substitute for oil futures prices and volatility can improve the accuracy of market analysis. This, in turn, can help policymakers in developing more efficient energy and economic policies by providing more precise insight into market dynamics. Research conducted by Liu et al. [68] and Lin et al. [69] demonstrates a growing trend towards using the decomposition prediction method. These developments highlight the need for governments to quickly adapt to evolving analytical techniques to ensure strong energy security and economic stability, as emphasized by Wang et al. [70] and Deng et al. [71]. The dynamic field of energy derivatives and more recent prediction models offer advantages and pose challenges to policymakers, requiring them to regularly review regulatory rules in order to capitalize on the advantages of new breakthroughs while minimizing possible hazards. Additional details on *Energy Derivative Markets* can be found in Appendix 1E.

3.6. Energy risk management

An aspect analyzed in the literature on energy risk management relates to electricity load forecasting. For example, Oreshin et al. [72] implemented a deep NN to analyze data from 35 European countries, showing that this technique outperforms ten classical baseline methods in both accuracy and forecast bias. Similarly, Moradzadeh et al. [73] used a cyber-secure deep learning framework that integrates Autoencoder, Convolutional Neural Network, and Long Short-Term Memory (LSTM) models on Iranian electricity load data, achieving the highest accuracy under both normal conditions and during a stealthy false data injection attack. Pelekis et al. [74] found that for Portuguese electricity data, the N-BEATS model consistently outperformed other models, with calendar and weather features like the hour of the day and temperature being key accuracy drivers. Furthermore, Zuniga-Garcia et al. [75] and Xu et al. [76] explored load forecasting in Mexico and France, respectively, with NN models providing accurate hourly load point forecasts. Finally, Haben et al. [77] review the literature on low voltage load forecasting, summarizing various case studies and methodologies from both AI and classical perspectives. Overall, the literature on electricity load forecasting suggests that implementing advanced NN models, such as deep learning frameworks and probabilistic predictive models, can significantly improve accuracy and reliability.

In energy management, various advanced techniques have been applied to optimize performance and reduce costs. Zou et al. [78] used a stochastic approach with unscented transform and reinforcement learning in multicarrier water and energy systems, significantly reducing operation and investment costs. Huang and Koroteev [79] applied a predictive analytic framework to urban waste production data, generating efficient waste models. Quiones et al. [80] utilized Encoder–Decoder Sequence-to-Sequence models with different RNN architectures for wind speed and solar radiation forecasting, identifying daily energy surpluses or shortfalls across seasons. Furthermore, Chien et al. [81] implemented an auto-executable blockchain-based peer-to-peer energy transaction market using LSTM NN, reducing electricity purchasing budgets by 17.68 % compared to traditional markets. Finally, Zhang et al. [82] demonstrated the high accuracy of various ANN models for short-, medium-, and long-term forecasting in the New England energy market.

Another direction of research focuses on energy markets and commodities. For example, Elmore and Dowling [83] used dynamic mode decomposition (DMD) and Augmented DMD (ADMD) for day-ahead market electricity prices in California, finding that ADMD is significantly faster and requires less data than traditional methods. Al-Alimi et al. [84] employed the enhancement of transformation reduction and long- and short-term memory integrated with ANN for energy markets like electricity and oil, achieving better and faster predictions. Amini and Khashei [85] used the empirical mode decomposition to forecast crude oil prices and wind power, outperforming other hybrid models. Fan et al. [86] applied a Markov decision process and LSTM model to predict natural gas demand and pricing, optimizing the pricing policy and outperforming other algorithms. Al-Musaylh et al. [87] used empirical wavelet transform models to analyze daily gas consumption in

Melbourne, achieving significant improvements over traditional methods. Additional details on the *topic* can be found in Appendix 1F.

3.7. Summary of the systematic literature review

The information summarized in Table 3 reveals several relevant patterns regarding both the structure of the literature and the role played by artificial intelligence across the main areas of energy finance. In four of the six research dimensions, namely Energy and Financial Markets, Pricing Mechanism, Energy Derivative Markets, and Energy Risk Management, the share of studies using AI methods is remarkably high, often exceeding ninety percent. This suggests a clear shift in these areas toward data-driven approaches, with many studies either replacing or augmenting traditional econometric models. The use of large-scale daily data and the need to respond to rapidly changing market dynamics have contributed to this transition. Across most pillars, the literature is dominated by empirical studies, particularly those focused on benchmarking or methodological innovation, while purely theoretical papers are relatively rare. Research with a policy-oriented focus is more common in areas such as Energy and Financial Markets, Energy Corporate Finance, and Green Finance and Investment, even though these fields still rely more heavily on classical econometric tools. Neural networks appear to be the most frequently used AI technique, while natural language processing remains less common. The focus on energy sectors also varies, with oil being predominant in Energy and Financial Markets, and electricity receiving greater attention in areas such as Pricing Mechanism and Risk Management. From a geographical perspective, the majority of studies concentrate on the European Union, the G7, and the BRICS countries, which reflects both data availability and the relevance of these economies for global energy trends. While econometric models remain central in domains like Energy Corporate Finance, their use is significantly reduced in other areas where AI techniques are increasingly dominant.

More to the point, across sections 3.1 to 3.6, the diffusion of AI techniques in energy finance is strongly shaped by the frequency and structure of the data, as well as by the type of decision-making involved. In section 3.1, approximately 65 % of the papers that employ AI are oriented toward short-term forecasting based on daily or intraday price series. These datasets are typically granular, stable, and systematically recorded, offering a suitable environment for non-linear models such as LSTM, SVM, or ensemble-based methods. The methodological choice reflects a practical need for adaptive models in high-frequency settings, where predictive accuracy on short horizons is economically valuable. Section 3.6 confirms this pattern, with nearly 90 % of the studies using AI to address problems related to market risk, volatility exposure, or scenario generation under uncertainty. In both cases, the wide adoption of AI appears less as a preference for innovation and more as a response to data environments that favor automation, re-training, and pattern recognition.

Still, the role of AI in other areas covered in sections 3.3 and 3.4 is less consistent. These sections include studies on corporate financial strategy and green investment behavior, where data is sparser, policy-driven constraints matter, and structural elements dominate the analysis. In these contexts, the benefits of advanced machine learning are not always clearly articulated. Some papers introduce complex models without benchmarking against conventional econometric tools, nor do they clarify how the added complexity contributes to interpretability or robustness. The tendency to default to AI, without a clear theoretical or empirical rationale, raises questions about whether the methodology is fully adapted to the research problem. While AI has proven value in specific domains, its extension across all subfields of energy finance should be guided by evidence rather than convention.

4. Energy finance Quo Vadis

Understanding how different stakeholder groups engage with energy finance offers a valuable perspective on the practical application of artificial intelligence methods across the field. While the conceptual literature is commonly structured around six pillars, the priorities and roles of actors such as consumers, traders, policymakers, and private companies vary significantly. These differences shape not only the types of data used but also the modeling preferences and the interpretability requirements of analytical outputs. Fig. 1 illustrates how the empirical connections between pillars differ depending on the needs of each actor. The overlaps in topic, methodology, energy subsector, and regional focus reveal that theoretical boundaries often become less rigid in real-world application. This perspective helps explain why methodological preferences diverge so markedly across subfields.

For the general public and regulated service providers, concerns are closely tied to affordability, reliability, and predictability of energy supply. These groups are not primarily focused on profit or performance, but rather on the fairness and transparency of the policies that shape their daily lives. As a result, models used to inform public pricing, subsidies, or consumption interventions must be interpretable and responsive to equity concerns. Explainable AI has the potential to increase the legitimacy of these decisions by clarifying how inputs such as income, household composition, or usage patterns influence final outcomes. A key question emerging from this context is how explainable models can be integrated into regulatory processes to enhance consumer trust. For example, one could explore whether incorporating explainable models into tariff adjustment mechanisms improves public acceptance and perceived fairness across income levels.

Traders operate in fundamentally different conditions. Their success depends on responding quickly to short-term fluctuations in prices, often using high-frequency data and predictive models that are optimized for performance rather than transparency. These actors routinely deploy neural networks and ensemble methods to anticipate shifts in the oil, gas, or electricity markets. In this setting, the focus is on speed and accuracy rather than interpretability. However, even in this environment, causal and explainable tools may play a useful role. A concrete research direction would be to examine whether traders can improve their risk exposure strategies by incorporating explainable components into black-box prediction pipelines, especially under uncertain regulatory conditions. Another open question is whether causal ML can help distinguish between signal and noise in trader reactions to policy announcements, or to better capture the level of information asymmetry during times of increased uncertainty.

Private companies, particularly those involved in infrastructure investment, energy production, or large-scale distribution, must navigate a hybrid space. Their decisions must satisfy financial objectives while also meeting ESG targets and aligning with evolving policy frameworks. While many firms already use AI for demand forecasting and operational optimization, they often lack tools that explain the reasoning behind outputs or quantify how policy interventions affect business outcomes. Causal methods can be particularly useful for evaluating, for instance, whether a firm's capital reallocation was truly a response to green financing incentives or merely coincided with a broader market trend. A relevant research question in this space would be whether firms that use causal analysis frameworks can better anticipate the financial impact of ESG-driven policy shifts and adapt their investment strategies accordingly. In parallel, explainable AI could support internal accountability and communication with stakeholders by revealing the key drivers behind energy-related financial performance indicators.

Government institutions face a distinct set of challenges that center on long-term policy goals. Their mandate includes ensuring climate resilience, energy security, and inclusive access, often through interventions that must be both targeted and socially acceptable. Traditional machine learning tools can forecast outcomes, but they cannot answer the question of whether a specific policy caused a specific effect. Causal machine learning allows for precisely this kind of inquiry. For example, during the 2021 European energy crisis, several governments implemented energy price caps intended to protect consumers rather

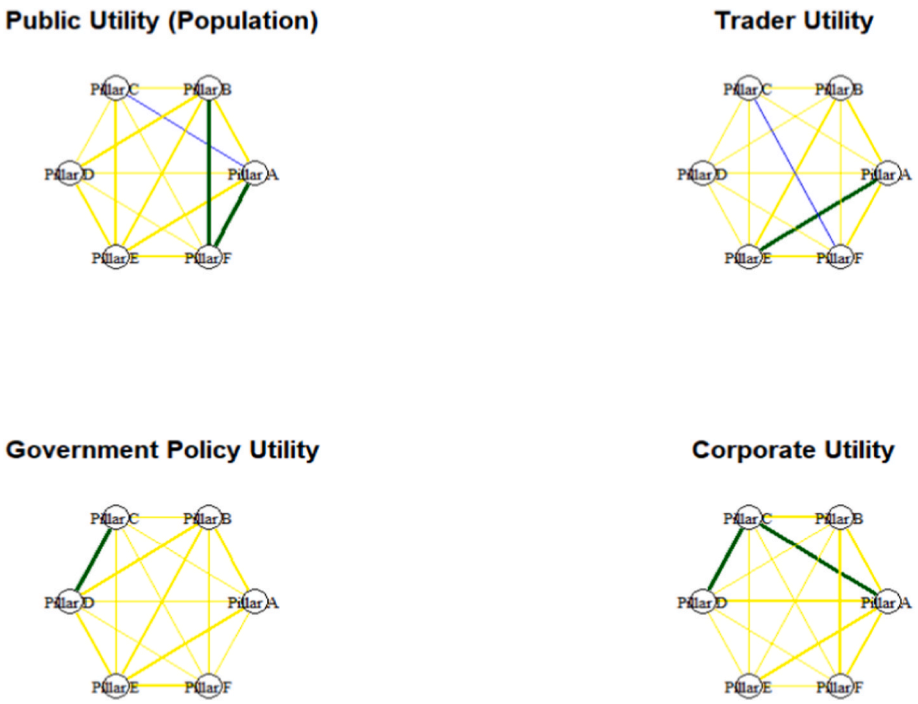


Fig. 1. The connection between pillars based on the utility of each stakeholder
Note: The connections between the pillars shown in the above graphs highlight the degree of interrelation based on common factors such as AI usage, applied methods, energy sectors, and geographic regions relative to the needs of different stakeholder categories (population, traders, governments, and corporations). Dark green lines show strong connections, yellow lines indicate moderate overlap in focus or methods, and blue lines represent weaker links with minimal interaction.
Source: own processing.

than stabilize markets. France froze electricity tariffs, Spain and Portugal introduced gas price caps through the Iberian mechanism, and other countries pursued similar approaches. These policy actions present valuable opportunities for causal investigation. Researchers could ask to what extent capped tariffs preserved household consumption, reduced financial distress, or prevented energy poverty. Moreover, explainable AI can complement this analysis by identifying which groups benefited most and whether targeting could be improved in future interventions.

Despite a growing number of studies using causal learning in environmental or macroeconomic contexts, relatively few investigate how energy policy affects household welfare, financial vulnerability, or inequality. This gap limits our understanding of how interventions resonate at the microeconomic level. A promising research direction would be to evaluate the causal impact of consumer-facing energy policies not only on average outcomes but also across income deciles, vulnerable groups, or regional contexts. For example, did green home subsidies reduce energy costs disproportionately for higher-income households, or were they effective in reaching low-income energy users? By linking such analysis with explainable AI, we can also begin to understand the mechanisms behind these distributional effects and offer more precise guidance to policymakers.

Another promising research direction concerns the profitability of battery energy storage systems operating through electricity arbitrage, particularly in relation to the role of policy support. A growing body of literature suggests that under standard market conditions, arbitrage alone does not typically generate sufficient returns to justify large-scale investment. Across different regional contexts, studies converge on the idea that profitability tends to depend on a combination of factors including market price volatility, technological progress, operational efficiency, and the existence of public incentives such as subsidies or capacity remuneration mechanisms [88–92]. This topic provides a suitable ground for the application of causal machine learning, which can be used to determine whether observed investments in battery systems are genuinely driven by these incentives or if they follow broader market dynamics. A key question is whether firms would have made the same investment decisions in the absence of targeted policy support. At the same time, explainable artificial intelligence can help identify the technical, regulatory, and financial variables that most strongly influence expected returns. By combining these approaches, future research could provide valuable insights into the conditions under which battery arbitrage becomes economically viable and into how public interventions might be designed to improve market participation and long-term adoption.

Finally, another underexplored dimension lies in distinguishing actual policy effects from expectations or sentiment-driven responses. This question is especially relevant for traders and firms operating in markets shaped by regulation. In domains such as carbon pricing or ESG-linked investment, it remains unclear whether behavioral shifts result from implemented measures or from anticipatory market reactions. For instance, Khan et al. [93] suggest that ESG performance is increasingly shaped by stakeholder pressure and institutional expectations, often preceding actual policy enforcement. This dynamic interplay between regulatory anticipation and corporate adaptation raises important challenges for impact attribution. Future research could use causal inference tools and explainable AI to disentangle the effects of announced versus implemented ESG policies, particularly in volatile markets.

5. Conclusions

This paper reviewed the recent literature on energy finance using the six-pillar framework originally proposed by Zhang et al. [1]. The review was based approximatively 700 articles retrieved from the Web of Science database, selected through a multistep filtering process consistent with the PRISMA protocol. Our objective was to assess how the field has evolved in recent years and how the adoption of artificial intelligence methods has influenced that evolution, both in terms of methodological orientation and practical application.

The findings show that artificial intelligence has become a central methodological tool in several core areas of energy finance. In pillars such as Energy and Financial Markets, Pricing Mechanism, Energy Derivative Markets, and Energy Risk Management, AI-based approaches now dominate the literature. Most of these studies focus on daily data and rely heavily on neural networks or other predictive techniques to anticipate price movements and manage risk. Benchmarking remains widespread, and the empirical focus is often placed on oil or electricity markets, particularly within the EU, G7, and BRICS regions. These areas reflect not only the maturity of data availability but also the responsiveness of market actors, especially traders and institutional investors, to new forecasting techniques.

Despite this rapid growth, the review highlights a notable methodological gap. Recent developments such as explainable artificial intelligence and causal machine learning remain largely absent across all six pillars. This absence is particularly evident in areas where policy relevance and transparency are essential. In fields such as Pricing Mechanism and Energy Risk Management, the lack of causal inference methods limits our ability to assess whether interventions such as price caps, subsidies, or environmental taxes actually achieved their intended effects. Similarly, the near-absence of explainable AI reduces the interpretability of model-based decisions, which is especially problematic when models are used in contexts involving public accountability or regulatory compliance. These observations suggest that the field could benefit from a more deliberate integration of these emerging methods into research designs that prioritize not only prediction, but also explanation and policy insight.

The review also shows that while the use of AI has increased, the nature of the data typically used in energy finance imposes practical constraints on model selection. Most datasets are structured and updated regularly but remain relatively limited in size and complexity. They usually consist of a few thousand daily observations and are often focused on price data, volatility indicators, or market spillovers. In this context, the consistent use of models such as shallow neural networks, support vector machines, or tree-based ensembles appears appropriate and has proven effective in many forecasting applications. However, the adoption of more elaborate architectures, such as those built for deep learning or sequential reasoning, is often difficult to justify given the available data. This mismatch becomes especially apparent in areas like corporate finance or sustainable investment, where data are more heterogeneous, less frequent, and shaped by long-term structural dynamics. In several instances, the use of complex models appears disconnected from the research question or policy relevance. This underscores the need for a better alignment between the complexity of the methods and the structure of the data, as well as a more critical reflection on when methodological novelty adds real analytical value.

This analysis can bring added value beyond the academic community. For researchers, this analysis provides a clear and structured understanding of how AI is currently used in energy finance, what benefits it brings in terms of price forecasting, operational efficiency and adaptability to energy market disruptions caused by global crises such as pandemics and geopolitical conflicts, and the limitations of such models. For practitioners, especially energy market traders and institutional investors, it highlights how AI models have been successfully used to improve short-term price forecasting, identify arbitrage opportunities more accurately and develop dynamic hedging strategies. For energy companies, AI applications in load forecasting and demand response contribute to better cost minimization and operational planning. For policymakers and regulators, the analysis draws attention to the limited use of interpretable models in areas where transparency is critical and shows how explainable artificial intelligence and causal machine learning could help close regulatory gaps, reduce information asymmetries, and improve the allocation and distribution of energy resources.

Declaration

During the preparation of this work, the authors used ChatGPT-4 and QuillBot to assist with language refinement and formatting of references. After using these tools, the authors thoroughly reviewed, edited, and validated all content to ensure clarity, accuracy, and consistency. The authors take full responsibility for the content of the final manuscript.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.esr.2025.101833.

Appendices

Appendix 1A. Energy and Financial Markets

Section	Authors, Year	Main contribution (yes/no)			Target (yes/no)			Data related aspects				Econometric techniques	AI techniques used (yes/no)			Paper also considers(yes/no)		
Energy and Financial Markets		theoretical	Methodology/ benchmarking	policy	oil	gas	electricity	Data type	Eu or G7	BRICS	Other	(yes/no)	AI/ ML	NN/ ANN	NLP	Causality	probabilistic forecasting	XAI
1	Ambrois et al. [94]	no	yes	yes	no	no	yes	Annual	yes	no	no	yes	yes	no	no	no	no	no
2	Ben Jabeur et al. [95]	no	yes	no	yes	no	no	Daily	yes	no	no	yes	yes	yes	no	yes	no	yes
3	Benatia [96]	no	no	yes	no	no	yes	Daily, hourly	yes	no	no	yes	yes	yes	no	no	yes	no
4	Chen [97]	no	no	yes	yes	no	no	Daily	yes	yes	yes	yes	yes	no	no	no	no	no
5	Cheng et al. [98]	no	yes	no	yes	no	no	Daily	yes	no	no	yes	yes	yes	no	yes	no	no
6	Das et al. [99]	no	yes	no	yes	no	no	Daily	yes	no	no	yes	yes	yes	no	yes	yes	no
7	Deng et al. [100]	no	yes	yes	yes	no	yes	Daily	yes	no	no	yes	yes	yes	no	yes	no	no
8	Dong et al. [41]	no	yes	yes	no	no	no	Daily	yes	no	no	yes	yes	yes	no	no	no	no
9	Düsterhöft et al. [101]	no	no	yes	yes	yes	yes	Panel data	yes	yes	yes	yes	yes	no	yes	no	no	no
10	Elamer et al. [102]	no	yes	yes	no	no	no	Daily	no	no	yes	yes	yes	yes	no	yes	no	no
11	Ghaemi Asl et al. [103]	no	yes	yes	yes	no	no	Daily	no	no	yes	yes	yes	yes	no	yes	no	No
12	Gupta [104]	no	no	yes	yes	no	no	Weekly	yes	no	no	yes	yes	no	no	no	yes	no
13	Jiang et al. [105]	no	yes	yes	no	no	no	Cross-section	no	yes	no	yes	yes	yes	no	no	no	no
14	Li et al. [14]	no	yes	yes	yes	no	no	Daily	yes	no	no	yes	yes	yes	yes	no	no	no
15	Lohrmann [106]	no	yes	yes	no	no	yes	Daily	yes	no	no	yes	yes	no	no	no	no	no
16	Loutfy [12]	no	yes	yes	yes	no	yes	Daily	yes	no	no	yes	yes	yes	yes	no	no	no
17	Lu et al. [8]	no	yes	yes	yes	no	no	Monthly	yes	no	no	yes	yes	yes	no	no	no	no
18	Lyu et al. [107]	no	yes	yes	yes	no	no	Monthly	yes	no	no	yes	yes	no	no	yes	no	no
19	Magazzino and Haroon [108]	no	yes	yes	no	no	yes	Monthly	yes	no	yes	yes	yes	yes	no	no	no	no
20	Mercadal [109]	yes	yes	no	no	no	yes	Hourly	yes	no	no	yes	yes	no	no	no	no	no
21	Mohsin [13]	no	yes	yes	yes	no	no	Daily	no	no	yes	yes	yes	yes	no	yes	no	no
22	Saadaoui [110]	no	yes	yes	no	no	yes	Daily	yes	no	yes	yes	yes	yes	no	no	no	no
23	Sadorsky [6]	no	yes	no	no	no	yes	Daily	yes	no	no	yes	yes	no	no	no	yes	no
24	Shen et al. [111]	no	no	yes	no	no	no	Daily	no	yes	no	yes	yes	no	no	yes	no	no
25	Song et al. [112]	no	yes	yes	yes	no	no	Daily	yes	no	no	yes	yes	no	no	yes	no	no
26	Wagner et al. [113]	no	yes	no	no	no	yes	Hourly	yes	no	no	yes	yes	yes	no	no	no	no
27	Wang and Wang [114]	no	yes	no	yes	no	no	Daily	yes	yes	no	yes	no	no	no	yes	no	no
28	Wang et al. [115]	no	yes	yes	yes	no	yes	Monthly	no	yes	no	yes	yes	yes	no	yes	no	no
29	Wang et al. [116]	no	yes	no	yes	no	no	Monthly	yes	yes	no	yes	Yes	no	no	yes	no	no
30	Wang et al. [43]	no	yes	yes	no	no	no	Monthly	no	yes	no	yes	yes	yes	no	yes	no	no
31	Wang, ZJ; Zhao, LT [9]	no	yes	no	yes	yes	yes	Daily	yes	no	no	yes	yes	no	no	no	no	no
32	Xu and Wang [10]	no	yes	no	yes	no	no	Monthly	no	no	yes	yes	yes	yes	no	no	yes	no
33	Xu et al. [45]	no	yes	yes	yes	no	no	Daily	yes	no	no	yes	yes	yes	no	yes	no	no
34	Yin and Wang [117]	no	yes	yes	yes	no	no	Monthly	yes	no	no	yes	yes	yes	no	yes	no	no
35	Yin et al. [118]	no	yes	yes	no	no	yes	Annual	no	no	yes	yes	yes	yes	no	no	no	no
36	Zhang et al. [7]	no	yes	no	yes	no	no	Daily	yes	no	no	yes	yes	no	no	yes	no	no
37	Zhang et al. [119]	no	yes	yes	yes	no	no	Daily	no	yes	no	yes	yes	no	no	no	no	no
38	Zhao et al. [120]	no	yes	yes	yes	no	no	Daily	yes	no	no	yes	yes	yes	no	yes	no	no
39	Zhou et al. [121]	no	no	yes	no	no	no	Daily	yes	no	no	yes	yes	yes	no	no	no	no
40	Zou [122]	no	yes	yes	yes	no	no	Daily	yes	no	no	yes	yes	yes	no	yes	no	no

Appendix 1B. Pricing Mechanisms

Section	Authors, Year	Main contribution (yes/no)			Target (yes/no)			Data related aspects				Econometric techniques	AI techniques used (yes/no)			Paper also considers (yes/no)		
Pricing Mechanisms		theoretical	Methodology/ benchmarking	policy	oil	gas	electricity	Data type	Eu or G7	BRICS	Other	(yes/no)	AI/ ML	NN/ ANN	NLP	Causality	probabilistic forecasting	XAI
1.	Alova [22]	no	no	yes	no	yes	no	Yearly	yes	yes	yes	yes	yes	yes	no	no	no	yes
2	Chellamani [17]	no	yes	no	no	no	yes	Hourly	no	no	no	no	yes	no	No	no	no	no
3	Cheng et al. [98]	no	yes	no	yes	no	no	Daily	yes	no	no	yes	yes	yes	no	yes	no	no
4	Dong et al. [123]	no	yes	yes	no	no	yes	10-min	no	yes	no	yes	yes	yes	no	no	no	no
5	Enrich et al. [23]	no	yes	yes	no	no	yes	Hourly	yes	no	no	yes	yes	no	no	yes	no	no
6	Hou et al. [124]	no	yes	no	no	No	yes	Hourly	yes	no	no	no	yes	no	no	no	no	no
7	Jiang et al. [125]	no	yes	yes	no	no	yes	10-min	no	yes	no	yes	yes	yes	no	no	no	no
8	Kim [16]	no	yes	no	no	no	yes	5-min	no	no	yes	no	yes	yes	no	no	no	no
9	Lee [126]	yes	yes	no	no	no	yes	Hourly	no	no	yes	no	yes	yes	no	no	no	no
10	Lee [18]	no	yes	no	no	no	yes	Hourly	no	no	no	no	yes	yes	no	no	no	no
11	Li et al. [25]	no	yes	yes	no	no	no	Daily	yes	no	no	yes	yes	yes	no	yes	yes	no
12	Lin et al. [127]	no	yes	no	no	yes	no	Daily	no	yes	no	no	yes	yes	no	no	no	no
13	Salazar et al. [19]	no	yes	no	no	no	yes	Hourly	no	no	yes	no	yes	no	no	no	no	no
14	Soysal [128]	no	yes	yes	no	no	yes	Hourly	yes	no	no	no	yes	yes	no	no	no	no
15	Talluri et al. [15]	no	yes	yes	no	no	yes	Panel data	yes	no	no	no	yes	yes	no	no	no	no
16	Wang et al. [44]	no	yes	no	yes	yes	yes	daily	no	yes	no	no	yes	yes	no	no	no	no
17	Wen et al. [20]	no	yes	no	no	no	yes	Weekly	no	yes	no	no	yes	yes	no	no	no	no
18	Yahsi et al. [129]	no	yes	no	no	no	no	Daily	yes	no	no	no	yes	yes	no	no	no	no
19	Zhao and Song [130]	no	yes	yes	no	no	yes	Quarter	no	yes	no	no	yes	yes	no	no	no	no
20	Zhao et al. [21]	no	yes	yes	no	no	yes	30-min	yes	no	no	no	yes	yes	no	no	no	no

Appendix 1C. Energy Corporate Finance

Section	Authors, Year	Main contribution (yes/no)			Target (yes/no)			Data related aspects				Econometric techniques	AI techniques used (yes/no)			Paper also considers (yes/no)		
Energy Corporate Finance		theoretical	Methodology/ benchmarking	policy	oil	gas	electricity	Data type	Eu or G7	BRICS	Other	(yes/no)	AI/ ML	NN/ ANN	NLP	Causality	probabilistic forecasting	XAI
1.	Anwer et al. [131]	no	yes	yes	NA	NA	NA	Yearly	yes	yes	no	yes	no	no	no	yes	no	no
2.	Apergis [35]	no	yes	yes	NA	NA	NA	Yearly	yes	no	no	yes	no	no	no	yes	no	no
3.	Bu et al. [31]	no	yes	yes	NA	NA	NA	Yearly	no	yes	no	yes	no	no	no	yes	no	no
4.	Cao et al. [132]	no	yes	yes	NA	NA	NA	Yearly	yes	yes	no	yes	no	no	no	yes	no	no
5.	Chang et al. [133]	no	yes	yes	NA	NA	NA	Yearly	no	yes	no	yes	no	no	no	yes	no	no
6.	Deng et al. [100]	no	yes	yes	yes	no	yes	Daily	yes	no	no	yes	yes	yes	no	yes	no	no
7.	Fonseka et al. [134]	no	yes	yes	NA	NA	NA	Yearly	no	yes	no	yes	no	no	no	yes	no	no
8.	Griffin et al. [135]	no	yes	yes	NA	NA	NA	Yearly	yes	no	no	yes	no	no	no	yes	no	no
9.	Guan et al. [136]	no	yes	yes	NA	NA	NA	Yearly	yes	no	yes	yes	no	no	no	yes	no	no
10.	Guo et al. [137]	no	yes	yes	NA	NA	NA	Yearly	no	yes	no	yes	no	no	no	yes	no	no
11.	Khiari et al. [138]	no	yes	yes	NA	NA	NA	Yearly	yes	no	no	yes	no	no	no	yes	no	no
12.	Kong et al. [30]	no	yes	yes	NA	NA	NA	Yearly	no	yes	no	yes	no	no	no	yes	no	no
13.	Lee and Xiao [29]	no	yes	yes	NA	NA	NA	Yearly	no	yes	no	yes	no	no	no	yes	no	no

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Section	Authors, Year	Main contribution (yes/no)			Target (yes/no)			Data related aspects				Econometric techniques	AI techniques used (yes/no)			Paper also considers (yes/no)		
Energy Corporate Finance		theoretical	Methodology/ benchmarking	policy	oil	gas	electricity	Data type	Eu or G7	BRICS	Other	(yes/no)	AI/ ML	NN/ ANN	NLP	Causality	probabilistic forecasting	XAI
14	Li [139]	no	yes	yes	NA	NA	NA	Yearly	no	yes	no	yes	no	no	no	yes	no	no
15	Li et al. [140]	no	yes	yes	NA	NA	NA	Yearly	no	yes	no	yes	no	no	no	yes	no	no
16.	**Li et al.** [141]	no	yes	no	no	yes	no	Daily & Weekly	yes	no	no	no	yes	yes	no	no	no	no
17.	Narayan and Nasiri [142]	no	yes	yes	NA	NA	NA	Yearly	yes	yes	yes	yes	no	no	no	yes	no	no
18.	Qiao et al. [143]	no	yes	yes	NA	NA	NA	Yearly	Yes	yes	yes	yes	no	no	no	yes	no	no
19.	Ren et al. [32]	no	yes	yes	NA	NA	NA	Yearly	no	yes	no	yes	no	no	no	yes	no	no
20.	Roy [144]	no	yes	yes	NA	NA	NA	Yearly	yes	no	no	yes	no	no	no	yes	no	no
21.	Shi [28]	no	yes	yes	NA	NA	NA	Yearly	no	yes	no	yes	no	no	no	yes	no	no
22.	Wan et al. [145]	no	yes	yes	NA	NA	NA	Yearly	yes	yes	yes	yes	no	no	no	yes	no	no
23.	Westerman et al. [146]	no	yes	yes	NA	NA	NA	Yearly	yes	no	no	yes	no	no	no	yes	no	no
24.	Wu and Huang [34]	no	yes	yes	NA	NA	NA	Yearly	no	yes	no	yes	no	no	no	yes	no	no
25.	Xu et al. [147]	no	yes	yes	NA	NA	NA	Yearly	no	yes	no	yes	no	no	no	yes	no	no
26.	Yin et al. [118]	no	yes	yes	no	no	yes	Annual	no	no	yes	yes	yes	yes	no	no	no	no
27.	Ying et al. [148]	no	yes	yes	NA	NA	NA	Yearly	no	yes	no	yes	no	no	no	yes	no	no
28.	Yu et al. [149]	no	yes	yes	NA	NA	NA	Yearly	no	yes	no	yes	no	no	no	yes	no	no
29.	Zhang and Kong [39]	no	yes	yes	NA	NA	NA	Yearly	no	yes	no	yes	no	no	no	yes	no	no

Appendix 1D. Green Finance and Investments

Section	Authors, Year	Main contribution (yes/no)			Target (yes/no)			Data related aspects				Econometric techniques	AI techniques used (yes/no)			Paper also considers (yes/no)		
Green Finance and Investments		theoretical	Methodology/ benchmarking	policy	oil	gas	electricity	Data type	Eu or G7	BRICS	Other	(yes/no)	AI/ ML	NN/ ANN	NLP	Causality	probabilistic forecasting	XAI
1.	Abrell et al. [150]	no	no	yes	yes*	yes*	yes	hourly	yes	no	no	no	yes	no	no	yes	yes	no
2	Abudu et al. [151]	no	yes	yes	no	yes	yes	text	NA	NA	NA	no	yes	yea	yes	no	no	no
3	Alabi et al. [152]	no	yes	yes	no	no	yes	hourly	yes	no	no	no	yes	yes	no	yes	yes	no
4	AL-Alimi et al. [84]	no	yes	yes	yes	no	Yes	High-frequency	no	no	yes	no	yes	yes	no	yes	yes	no
5	Aldy et al. [48]	no	no	yes	no	no	Yes	Review paper	yes	no	no	no	no	no	no	yes	no	no
6	Alova and Caldecott [22]	no	yes	yes	yes	yes	Yes	granular information	yes	yes	yes	yes	yes	no	no	yes	yes	no
7	Debrah et al. [50]	no	yes	yes	no	no	no	Review paper	–	–	–	no	no	no	no	no	no	no
8	Ding [59]	no	yes	yes	no	no	yes	hourly	yes	no	no	yes	yes	no	no	yes	no	no
9	Ding et al. [52]	no	yes	yes	yes	yes	yes	yearly	no	yes	no	yes	no	no	no	yes	no	no
10	Du et al. [54]	no	no	yes	no	no	no	yearly	no	yes	no	yes	no	no	no	yes	no	no
11	Fang et al. [153]	no	yes	yes	no	no	yes	annual	no	yes	no	yes	yes	yes	no	no	no	no
12	Fang et al. [58]	no	no	yes	no	no	no	Cost estimates for hydrogen production	no	yes	no	yes	no	no	no	yes	no	no
13	Fernández-Martínez and Jaramillo-Morán [154]	no	no	yes	no	no	yes	hourly	yes	no	no	no	yes	yes	no	yes	yes	no
14	Fisch-Romito et al. [155]	no	no	yes	yes	yes	yes	Review paper	yes	yes	yes	no	yes	no	no	no	no	no
15	Ghoddusi et al. [3]	no	yes	yes	yes	yes	yes	Review paper	yes	yes	yes	no	yes	yes	yes	no	no	no
16	Han and Yang [156]	no	yes	yes	no	no	yes	Yearly	no	yes	no	yes	yes	yes	no	yes	no	no

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Section	Authors, Year	Main contribution (yes/no)			Target (yes/no)			Data related aspects				Econometric techniques	AI techniques used (yes/no)			Paper also considers (yes/no)		
Green Finance and Investments		theoretical	Methodology/ benchmarking	policy	oil	gas	electricity	Data type	Eu or G7	BRICS	Other	(yes/no)	AI/ ML	NN/ ANN	NLP	Causality	probabilistic forecasting	XAI
17	Hou et al. [51]	no	no	yes	no	no	yes	cross-country panel data	yes	yes	yes	yes	no	no	no	yes	no	no
18	Huang et al. [62]	no	yes	yes	yes	yes	yes	daily	yes	no	no	yes	yes	yes	no	yes	yes	no
19	Ismer et al. [157]	no	no	yes	no	no	yes	–	yes	no	no	no	no	no	no	yes	no	no
20	Lei et al. [42]	no	yes	yes	yes	yes	yes	daily	no	yes	No	yes	yes	no	no	yes	yes	no
21	Li et al. [43]	no	yes	yes	yes	yes	yes		yes	no	no	yes	yes	yes	no	yes	yes	no
22	Luo and Oyedele [158]	no	yes	yes	no	yes	yes	monthly	no	yes	no	no	no	no	no	yes	yes	no
23	Martinez-Alvarez et al. [61]	no	no	yes	yes	yes	yes	yearly	yes	yes	yes	yes	no	no	no	yes	no	no
24	Meckling [49]	no	no	yes	yes	yes	yes	–	yes	no	no	no	no	no	no	no	no	no
25	Mohsin and Jamaani [13]	no	yes	yes	yes	no	no	daily	yes	yes	yes	yes	yes	yes	no	yes	yes	no
26	Nadirgil [46]	no	yes	yes	yes	yes	yes	daily/ quarterly	yes	no	no	yes	yes	no	no	yes	yes	no
27	Nie et al. [60]	no	yes	yes	no	no	yes	Quantitative text analysis methods	yes	yes	yes	no	no	no	no	no	no	no
28	Qiu et al. [159]	no	yes	yes	no	yes	yes	hourly	yes	no	no	no	yes	yes	no	no	no	no
29	Soltani [160]	no	yes	yes	no	no	yes	annual	yes	yes	yes	yes	yes	yes	no	no	no	no
30	Song et al. [55]	no	no	yes	no	no	no	yearly	no	yes	no	yes	no	no	no	yes	yes	no
31	Tan et al. [161]	no	yes	yes	no	no	yes	Yearly	no	yes	no	yes	yes	yes	yes	no	no	no
32	Wan et al. [162]	no	yes	yes	no	no	yes	hourly	–	–	–	no	no	no	no	yes	yes	no
33	Wang et al. [163]	no	yes	yes	yes	yes	no	NA	no	yes	no	no	yes	yes	no	no	yes	no
34	Wang et al. [44]	no	yes	no	yes	yes	yes	daily	no	yes	no	no	yes	yes	no	no	no	no
35	Wang et al. [70]	no	yes	yes	no	no	yes	hourly	no	no	yes	no	yes	yes	no	no	yes	no
36	Woody et al. [164]	no	no	yes	no	no	yes	hourly	yes	no	no	yes	no	no	no	no	no	no
37	Xiao et al. [47]	no	yes	no	yes	no	no	Monthly	yes	no	no	yes	yes	yes	no	no	no	no
38	Xu et al. [45]	no	yes	yes	yes	no	no	Daily	yes	no	no	yes	yes	yes	no	yes	no	no
39	Yang [57]	no	no	yes	no	no	yes	annual	no	yes	no	yes	no	no	no	no	no	no
40	Zhang et al. [56]	no	no	yes	no	no	yes	annual	no	yes	no	yes	no	no	no	yes	no	no
41	Zhao et al. [120]	no	yes	yes	yes	no	no	Daily	yes	no	no	yes	yes	yes	no	yes	no	no
42	Zhao et al. [165]	no	yes	yes	no	no	yes	annual	no	yes	no	yes	yes	yes	no	yes	no	no
43	Zhou et al. [166]	no	yes	yes	no	no	yes	annual	yes	yes	no	no	no	no	no	no	no	no
44	Zhou et al. [53]	no	no	yes	no	no	yes	annual	yes	yes	yes	no	no	no	no	no	no	no
45	Zhu et al. [167]	no	yes	yes	no	no	yes	annual	no	yes	no	yes	yes	yes	no	yes	no	no

Appendix 1E

Section	Authors, Year	Main contribution (yes/no)			Target (yes/no)			Data related aspects				Econometric techniques	AI techniques used (yes/no)			Paper also considers (yes/no)		
Energy Derivative Markets		theoretical	Methodology/ benchmarking	policy	oil	gas	electricity	Data type	Eu or G7	BRICS	Other	(yes/no)	AI/ ML	NN/ ANN	NLP	Causality	probabilistic forecasting	XAI
1	Butler et al. [168]	no	yes	no	yes	no	no	Daily	yes	no	no	yes	no	yes	no	no	no	no
2	Castello and Resta [169]	no	yes	no	no	yes	no	Daily	yes	no	no	no	no	yes	no	no	no	no

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Section	Authors, Year	Main contribution (yes/no)			Target (yes/no)			Data related aspects				Econometric techniques	AI techniques used (yes/no)			Paper also considers (yes/no)		
Energy Derivative Markets		theoretical	Methodology/ benchmarking	policy	oil	gas	electricity	Data type	Eu or G7	BRICS	Other	(yes/no)	AI/ ML	NN/ ANN	NLP	Causality	probabilistic forecasting	XAI
3	Creamer [66]	no	yes	no	yes	yes	no	Daily	yes	no	no	yes	yes	no	no	yes	no	no
4	Deng et al. [100]	no	yes	yes	yes	no	yes	Daily	yes	no	no	yes	yes	yes	no	yes	no	no
5	Deng et al. [170]	no	yes	no	yes	no	no	Daily	yes	yes	no	no	no	yes	no	no	no	no
6	Ding et al. [171]	no	yes	no	yes	no	yes	30 Minutes	no	yes	no	yes	yes	yes	no	no	no	no
7	Gong et al. [172]	no	yes	no	yes	no	no	Daily	yes	yes	no	no	yes	no	yes	no	no	yes
8	Guo et al. [173]	no	yes	yes	yes	no	no	Daily	no	yes	no	no	yes	yes	no	no	no	no
9	Hajiabotorabi et al. [174]	no	yes	no	yes	no	no	Daily	yes	no	no	yes	yes	yes	no	no	no	yes
10	Huang et al. [175]	no	yes	no	yes	no	no	Daily	no	yes	no	yes	no	yes	no	no	no	no
11	Huang et al. [176]	no	yes	mo	yes	no	no	Daily	no	yes	no	no	yes	yes	no	no	no	yes
12	Jana and Ghosh [177]	no	yes	no	no	yes	no	Daily	yes	no	no	no	yes	no	no	mo	no	yes
13	Jiang et al. [178]	no	yes	no	yes	yes	no	Daily	yes	no	no	yes	yes	yes	no	no	yes	no
14	Jiang et al. [40]	no	yes	no	yes	no	no	5 Minutes	no	yes	no	yes	no	yes	no	no	no	no
15	Kaplan et al. [179]	no	yes	no	yes	no	no	Daily	yes	no	no	no	no	yes	yes	no	no	no
16	Li and Li [180]	no	yes	no	yes	no	no	Weekly	yes	no	no	yes	yes	yes	no	no	no	no
17	Li and Wang [181]	no	yes	no	yes	yes	no	Daily	yes	no	no	no	no	yes	no	no	no	no
18	Li et al. [141]	no	yes	no	no	yes	no	Daily & Weekly	yes	no	no	no	yes	yes	no	no	no	no
19	Li et al. [182]	no	yes	no	yes	no	no	Daily	yes	no	no	yes	yes	yes	yes	no	no	no
20	Lin et al. [69]	no	yes	no	yes	no	no	Daily	yes	no	no	no	yes	yes	no	no	no	no
21	Liu et al. [183]	no	yes	no	yes	no	no	Daily	yes	no	no	no	no	yes	no	no	no	no
22	Liu et al. [68]	no	yes	no	yes	yes	no	Daily	yes	no	no	no	no	yes	no	no	no	no
23	Long et al. [184]	no	yes	no	yes	no	no	Daily & Weekly	yes	no	no	no	no	yes	no	no	yes	no
24	Lu et al. [8]	no	yes	yes	yes	no	no	Monthly	yes	no	no	yes	yes	yes	no	no	no	no
25	Luo et al. [153]	no	yes	no	yes	no	no	Daily	yes	no	no	yes	no	yes	no	no	no	no
26	Luo et al. [185]	no	yes	no	yes	no	no	5 Minutes	yes	no	no	yes	yes	no	no	no	no	yes
27	Lyu et al. [107]	no	yes	yes	yes	no	no	Monthly	yes	no	no	yes	yes	no	no	yes	no	no
28	Manickavasagam et al. [186]	no	yes	no	yes	no	no	Hourly	yes	no	no	no	no	yes	no	no	no	no
29	Shaikh [67]	no	yes	no	yes	no	no	Daily	yes	no	no	yes	no	yes	no	no	no	no
30	Su et al. [187]	no	yes	no	yes	no	no	Daily	no	no	yes	no	yes	yes	no	no	no	no
31	Sun et al. [188]	no	yes	no	yes	no	no	Daily	yes	no	no	no	no	yes	no	no	no	no
32	Sun et al. [189]	no	yes	no	yes	no	no	Daily	yes	no	no	yes	yes	yes	mo	no	yes	no
33	Verma [190]	no	yes	no	yes	no	no	Daily	yes	no	no	yes	no	yes	no	no	no	no
34	Wang and Wang [114]	no	yes	no	yes	no	no	Daily	yes	yes	no	yes	no	no	no	yes	no	no
35	Wang and Wang [191]	no	yes	no	yes	yes	no	Daily	yes	no	no	no	yes	yes	no	no	no	no
36	Wang and Wang [192]	no	yes	no	yes	no	no	Daily	yes	no	no	no	yes	yes	no	no	no	no
37	Wang and Wang [63]	no	yes	no	yes	yes	no	Daily	yes	no	no	yes	no	yes	no	no	no	no
38	Wang and Wang [64]	no	yes	no	yes	no	no	Daily	yes	no	no	no	no	yes	no	no	no	no
39	Wang et al. [193]	no	yes	no	yes	no	no	Daily	yes	yes	no	no	no	yes	no	no	no	no
40	Wang et al. [194]	no	yes	no	yes	no	no	Daily	yes	yes	no	no	yes	yes	yes	no	no	no
41	Weng et al. [195]	no	yes	no	yes	no	no	Daily	yes	no	no	yes	yes	yes	yes	no	no	no
42	Wu et al. [196]	no	yes	no	yes	no	no	Daily	yes	no	no	no	no	yes	no	no	no	no
43	Wu et al. [197]	no	yes	yes	yes	no	no	Daily	no	yes	no	no	no	yes	yes	no	no	no
44	Xiao et al. [47]	no	yes	no	yes	no	no	Monthly	yes	no	no	yes	yes	yes	no	no	no	no
45	Xu et al. [45]	no	yes	yes	yes	no	no	Daily	yes	no	no	yes	yes	yes	no	yes	no	no
46	Yang et al. [65]	no	yes	no	yes	yes	no	Daily	yes	no	no	no	yes	yes	no	no	no	no
47	Yin and Wang [198]	no	yes	no	yes	no	no	Daily	yes	no	no	yes	no	yes	no	no	no	yes
48	Zhang and Wang [199]	no	yes	no	yes	no	no	Daily	yes	yes	no	no	no	yes	no	no	no	no
49	Zhang et al. [119]	no	yes	yes	yes	no	no	Daily	no	yes	no	yes	yes	no	no	no	no	no
50	Zhao et al. [200]	no	yes	no	yes	no	no	Daily	yes	yes	no	yes	yes	yes	no	no	yes	no

Appendix 1F. Energy Risk Management

Section	Authors, Year	Main contribution (yes/no)			Target (yes/no)			Data related aspects				Econometric techniques	AI techniques used (yes/no)			Paper also considers (yes/no)		
Energy Risk Management		theoretical	Methodology/ benchmarking	policy	oil	gas	electricity	Data type	Eu or G7	BRICS	Other	(yes/no)	AI/ ML	NN/ ANN	NLP	Causality	probabilistic forecasting	XAI
1	Ahmad and Zhang [201]	no	yes	no	no	no	yes	hourly	yes	no	no	no	yes	yes	no	no	no	no
2	AL-Alimi et al. [84]	no	yes	no	yes	no	yes	15 min data	no	no	yes	yes	yes	yes	no	no	no	no
3	AL-Musaylh et al. [87]	no	yes	no	no	yes	no	Daily data	no	no	yes	yes	yes	no	no	no	no	no
4	Alshater et al. [202]	no	yes	no	no	no	no	Daily	yes	no	no	no	yes	yes	no	no	no	no
5	Amini et al. [85]	no	yes	no	yes	no	yes	Daily and hourly data	yes	no	no	yes	yes	no	no	no	no	no
6	Azam and Younis [203]	no	yes	no	no	no	yes	Hourly	yes	no	no	no	yes	yes	no	no	no	yes
7	Brusaferri et al. [204]	no	yes	no	no	no	yes	Hourly	yes	no	no	no	yes	yes	no	no	yes	no
8	Chien et al. [81]	yes	yes	no	no	no	yes	Hourly data	no	no	yes	yes	yes	no	no	no	yes	no
9	Chinmoy et al. [205]	A review paper that cannot be classified.																
10	Condemi et al. [206]	no	yes	yes	no	no	yes	Monthly	yes	no	no	yes	yes	yes	no	no	no	no
11	Das et al. [207]	no	yes	no	no	no	yes	Intraday	yes	no	no	yes	yes	yes	no	no	no	no
12	E et al. [208]	no	yes	no	yes	yes	no	Monthly, daily	yes	no	no	no	yes	yes	no	no	no	no
13	Elmore et al. [83]	no	yes	no	no	no	yes	Hourly	yes	no	no	yes	yes	no	no	no	no	no
14	Fan et al. [86]	no	yes	no	no	yes	no	Hourly	no	yes	no	yes	yes	yes	no	no	no	no
15	Ferrari et al. [27]	no	yes	no	yes	yes	no	Quarterly	yes	yes	yes	yes	yes	no	no	no	no	no
16	Ftiti et al. [209]	no	yes	yes	yes	yes	no	Daily	yes	no	no	yes	yes	no	no	no	no	no
17	Gheorghiu et al. [210]	no	yes	no	no	no	yes	15-min	yes	no	no	no	yes	yes	no	no	no	no
18	Haben et al. [77]	yes	no	yes	no	no	yes	Review paper	yes	yes	yes	yes	no	no	no	no	no	no
19	Herrera et al. [26]	no	yes	yes	yes	yes	no	Monthly	yes	yes	yes	yes	yes	yes	no	no	no	no
20	Huang and Koroteev [79]	no	yes	yes	no	no	no	Daily and weekly	no	yes	no	no	yes	no	no	no	no	no
21	Jahangir et al. [211]	no	yes	no	no	no	yes	hourly	yes	no	no	no	yes	yes	no	no	no	no
22	Jasiński [212]	no	yes	no	no	no	yes	hourly	yes	no	no	no	no	yes	no	no	no	no
23	Kaya et al. [213]	no	yes	no	no	no	yes	Hourly	no	no	yes	no	no	yes	no	no	no	no
24	Lin et al. [127]	no	yes	no	no	yes	no	Daily	no	yes	no	no	yes	yes	no	no	no	no
25	Magazzino and Haroon [108]	no	yes	yes	no	no	yes	Monthly	yes	no	yes	yes	yes	yes	no	no	no	no
26	Moradzadeh et al. [73]	no	yes	no	no	no	yes	Hourly	no	yes	no	yes	yes	yes	no	no	no	no
27	Oreshkin et al. [72]	no	yes	no	no	no	yes	Monthly	yes	no	yes	yes	yes	yes	no	no	no	no
28	Pelekis et al. [74]	no	yes	no	no	no	yes	Hourly data	yes	no	no	yes	yes	yes	no	no	no	no
29	Qin et al. [214]	no	yes	no	yes	yes	yes	Daily, monthly	yes	no	no	yes	yes	yes	no	no	no	no
30	Quiñones et al. [80]	no	yes	yes	no	no	yes	Hourly	yes	no	no	yes	yes	yes	no	no	no	no
31	Saadaoui [110]	no	yes	yes	no	no	yes	Daily	yes	no	yes	yes	yes	yes	no	no	no	no
32	Suarez-Cetrulo et al. [215]	no	yes	no	no	no	yes	Hourly	yes	no	no	yes	yes	yes	no	no	no	no
33	Taghikhan et al. [216]	no	yes	yes	no	no	yes	Survey	no	no	yes	yes	yes	yes	no	no	no	no
34	Wang and Wang [114]	no	yes	no	yes	no	no	Daily	yes	yes	no	yes	no	no	no	yes	no	no
35	Wang and Wang [63]	no	yes	no	yes	yes	no	Daily	yes	no	no	yes	no	yes	no	no	no	no

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Section	Authors, Year	Main contribution (yes/no)			Target (yes/no)			Data related aspects				Econometric techniques	AI techniques used (yes/no)			Paper also considers (yes/no)		
Energy Risk Management		theoretical	Methodology/ benchmarking	policy	oil	gas	electricity	Data type	Eu or G7	BRICS	Other	(yes/no)	AI/ ML	NN/ ANN	NLP	Causality	probabilistic forecasting	XAI
36	Xu et al. [45]	no	yes	yes	yes	no	no	Daily	yes	no	no	yes	yes	yes	no	yes	no	no
37	Xu et al. [76]	no	yes	no	no	no	yes	Half-hourly data	yes	no	no	yes	yes	no	no	no	yes	no
38	Yang and Schell [217]	no	yes	no	no	no	yes	Intraday (5min)	yes	no	no	no	yes	yes	no	no	no	no
39	Zhang and Wang [199]	no	yes	no	yes	no	no	Daily	yes	yes	no	no	no	yes	no	no	no	no
40	Zhang et al. [218]	no	yes	yes	no	no	yes	Monthly	no	yes	no	yes	yes	yes	no	yes	no	yes
41	Zhang et al. [82]	no	yes	yes	no	no	yes	Daily data	no	no	yes	yes	yes	yes	no	no	no	no
42	Zhao et al. [219]	no	yes	no	yes	no	no	Monthly	no	yes	no	yes	yes	no	yes	yes	no	no
43	Zuniga-Garcia et al. [75]	no	yes	no	no	no	yes	15 min data	no	no	yes	yes	yes	yes	no	no	yes	no

Data availability

Data will be made available on request.

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